

Design of an Operational Amplifier and Its Use in an LDO with Bandgap Reference

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Abstract—This report presents the schematic design and simulation of an operational amplifier, a bandgap reference, and a low-dropout regulator (LDO). The circuits were implemented in the 180 nm high-voltage CMOS process using Cadence Virtuoso. The operational amplifier was first designed and tested as a separate block. It was then used as the error amplifier in the LDO. The bandgap circuit provides the reference voltage for the regulator. The op-amp was tested for DC gain, unity-gain bandwidth, slew rate, CMRR, input common-mode range, output swing, and current consumption. The bandgap reference was simulated over process, supply-voltage, and temperature variations. Its startup operation was also verified during a supply ramp. The output voltage changed from about 0.993 V to 1.007 V. This gives a total variation of about 1.4%. The LDO produced about 1.794 V under nominal conditions. It was tested for DC regulation, stability, PSRR, current consumption, and transient load response. The phase margin was above 60° under regulated operating conditions above the dropout boundary. Possible improvements to the transient response are also discussed.

Index Terms—Operational amplifier, bandgap reference, LDO, low-dropout regulator, analog integrated circuits, XFAB XH018, Cadence Virtuoso, PVT simulation.

I. INTRODUCTION

Operational amplifiers, bandgap references, and low-dropout regulators are common building blocks in analog integrated circuits. The op-amp provides gain and controls the feedback loop. The bandgap reference generates a voltage that is relatively insensitive to supply and temperature changes. The LDO uses these blocks to produce a stable output voltage from an unregulated supply.

In this project, the three circuits were combined into a complete regulator system. The bandgap reference generates a voltage close to 1 V. The LDO feedback network senses and scales the output voltage. The op-amp compares this feedback signal with the reference voltage and drives the gate of a PMOS pass transistor. The target LDO output is approximately 1.8 V. The op-amp and bandgap reference were also tested separately before they were used in the complete LDO.

The circuits were implemented in the 180 nm high-voltage CMOS process using Cadence Virtuoso and Spectre. Simulations covered the TT, FF, SS, FS, and SF process corners. The supply voltage was varied from 1.8 V to 3.3 V, and the temperature range was -40°C to 85°C . The main challenges were low current consumption, sufficient op-amp gain and

bandwidth, and stable LDO operation. The instantaneous load-current requirement was another major limitation.

II. SYSTEM-LEVEL ARCHITECTURE

The complete system consists of three main blocks: the bandgap reference, the operational amplifier, and the LDO output stage. The bandgap reference generates an approximately 1 V reference voltage and also provides the bias current used by the operational amplifier. The operational amplifier compares this reference voltage with the feedback voltage taken from the LDO output.

The LDO output stage is formed by a PMOS pass transistor, a resistive feedback divider, the load capacitor, and the load-current source. The op-amp output drives the PMOS gate and adjusts the pass current according to the feedback error. In this way, the bandgap reference, op-amp, and PMOS pass device operate together as a closed-loop voltage regulator.

The detailed operation and sizing of each block are discussed separately in the following sections.

III. OPERATIONAL AMPLIFIER DESIGN

The operational amplifier was designed as a two stage CMOS amplifier with an NMOS differential input pair, a PMOS current mirror active load, a second common source gain stage, and Miller compensation. This topology was selected because it provides sufficient voltage gain with a moderate current budget and can be compensated for stable operation both as a stand alone amplifier and as the LDO error amplifier.

The first stage of the amplifier is an NMOS differential pair with a PMOS current mirror active load. The differential pair converts the input voltage difference into signal current, while the active load performs differential to single ended conversion and provides high small signal resistance at the first stage output node. This structure gives the amplifier its first voltage gain contribution without using large passive resistors.

The second stage is implemented as a common source gain stage biased by a current source device. This stage increases the total DC gain and provides the required output voltage swing for driving the following circuitry. In the LDO application, this output node directly drives the PMOS pass transistor gate. Therefore, the output stage must be able to

move over a sufficiently wide voltage range to increase or decrease the pass-device current.

The amplifier uses Miller compensation between the first stage output node and the final output node. A series compensation resistor is included with the compensation capacitor. The compensation capacitor creates a dominant pole and improves the stability of the two stage amplifier. The series resistor helps control the zero introduced by the compensation path and improves the phase response.

This compensation network was also important after the operational amplifier was inserted into the LDO loop. The PMOS pass device and the load capacitor introduce additional poles in the LDO system. Therefore, the compensation resistor was adjusted during LDO simulations to reduce the loop crossover frequency and improve phase margin. With the final compensation setting, the LDO achieved phase margin above 60° for regulated supply cases above the dropout boundary.

A. Biasing and Current Budget

The operational amplifier uses a reference current input, I_{ref} , to set its internal bias currents. In the stand alone operational amplifier simulations, this current was applied externally to characterize the amplifier independently. In the integrated LDO simulations, I_{ref} is supplied by the bandgap reference circuit, so the bandgap provides both the reference voltage and the bias current for the final system.

The internal branch currents are generated using MOS current mirrors. The current mirror ratios and transistor dimensions were selected to keep the total op-amp current below the $100\mu\text{A}$ limit while still providing sufficient gain, bandwidth, and slew-rate capability. This creates a tradeoff between low-power operation and speed.

The total op-amp supply current was verified across all simulated PVT cases. As shown in Fig. 1, the current remains below the $100\mu\text{A}$ specification in every case. The simulated current ranges from approximately $79\mu\text{A}$ to $94\mu\text{A}$.

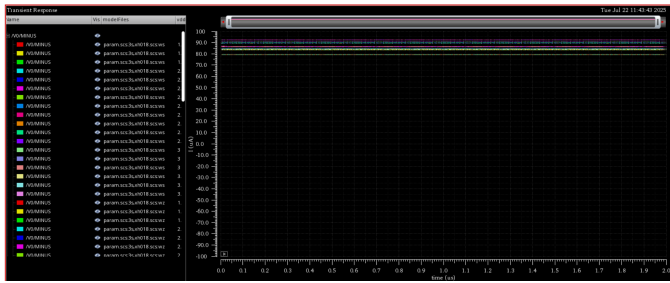


Fig. 1. DC current of the operational amplifier across the simulated PVT corners.

The DC operating point of the operational amplifier was also checked after the final sizing. Fig. 2 shows a representative DC operating-point annotation at $V_{DD} = 3.3\text{ V}$. The transistor terminal voltages, drain currents, and small-signal parameters were inspected to confirm that the devices operate in the intended regions. This check was important because the op-amp must preserve its gain, input/output range, and biasing

condition without entering undesired linear or subthreshold operation.

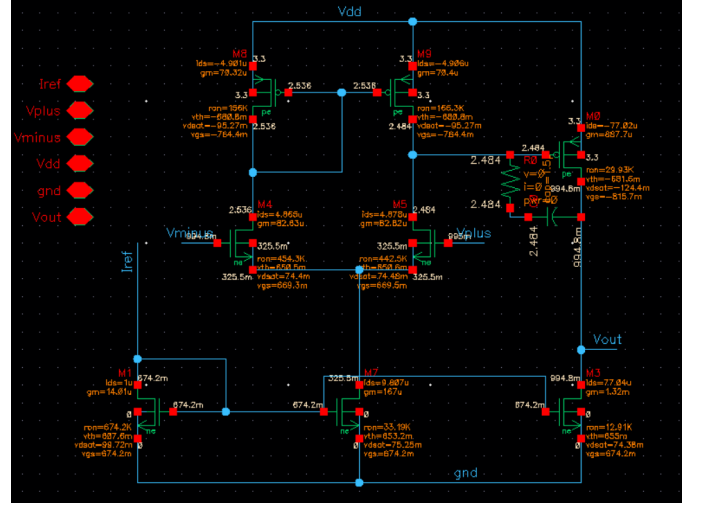


Fig. 2. DC operating point annotation of the operational amplifier at $V_{DD} = 3.3\text{ V}$.

B. DC Gain

The DC gain of the operational amplifier is provided by the cascaded gain of the differential input stage and the second common-source stage. The first stage provides differential sensing and initial voltage gain, while the second stage increases the overall open-loop gain at the output node. High small-signal output resistance in both stages helps increase the DC gain.

The open-loop gain was simulated across process, voltage, and temperature corners. As shown in Fig. 3, the nominal DC gain is approximately 100 dB. Across all simulated cases, the minimum gain is 97.34 dB, obtained at $V_{DD} = 3.3\text{ V}$, 85°C , SS corner. The maximum gain is 105.16 dB, obtained at $V_{DD} = 2.7\text{ V}$, -40°C , SF corner. Therefore, the op-amp satisfies the required DC gain specification of more than 40 dB with a large margin.

TABLE I
OPERATIONAL AMPLIFIER DC GAIN SUMMARY

Condition	DC Gain	Comment
Nominal case	$\approx 100\text{ dB}$	Typical operation
Worst case	97.34 dB	$V_{DD} = 3.3\text{ V}$, 85°C , SS
Best case	105.16 dB	$V_{DD} = 2.7\text{ V}$, -40°C , SF

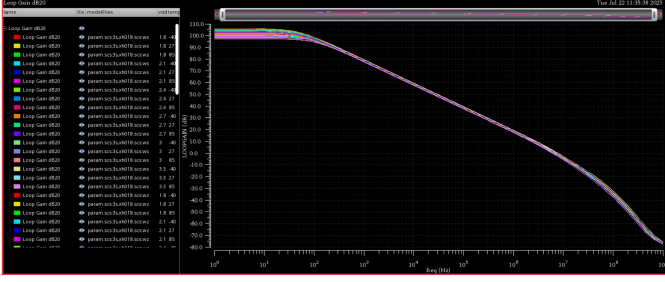


Fig. 3. Open-loop DC gain of the operational amplifier across PVT corners.

C. Unity-Gain Bandwidth

The unity-gain bandwidth of the operational amplifier was verified using AC simulations across all process, voltage, and temperature cases. The unity gain bandwidth remains higher than the required 5 MHz specification in every simulated case. The minimum unity-gain bandwidth is 5.787 MHz, obtained at $V_{DD} = 1.8\text{ V}$, 85°C , SS corner. The maximum value is 7.646 MHz, obtained at $V_{DD} = 2.7\text{ V}$, -40°C , FF corner.

TABLE II
OPERATIONAL AMPLIFIER UNITY-GAIN BANDWIDTH SUMMARY

Condition	UGB	Comment
Minimum case	5.787 MHz	$V_{DD} = 1.8\text{ V}$, 85°C , SS
Maximum case	7.646 MHz	$V_{DD} = 2.7\text{ V}$, -40°C , FF

D. Slew Rate

The slew rate was verified using a unity-gain buffer testbench. A pulse input was applied to the amplifier, and the output transient response was observed. The slew rate was calculated from the slope of the output waveform during the rising and falling transitions as

$$SR = \frac{\Delta V_{OUT}}{\Delta t}. \quad (1)$$

The simulation results show that the slew rate does not change significantly across process, voltage, and temperature corners. Due to the NMOS input pair based two stage topology and the selected output-stage biasing, the positive output transition is faster than the negative transition. If a PMOS input pair and the opposite output-stage polarity were used, the dominant fast and slow transition directions would be expected to be reversed. In the final design, the rising slew rate is approximately $10.62\text{ V}\mu\text{s}^{-1}$, while the falling slew rate is approximately $5.45\text{ V}\mu\text{s}^{-1}$. Therefore, the rising transition satisfies the $8\text{ V}\mu\text{s}^{-1}$ specification, while the falling transition is slower and should be reported as the limiting slew-rate case. Although the falling slew rate is lower than the $8\text{ V}\mu\text{s}^{-1}$ target, this design point was kept because the amplifier satisfies the other major op-amp requirements with good margin. In particular, DC operating-point simulations showed that the transistors remain in the intended saturation region across the verified input and output ranges, without entering linear or subthreshold operation. Therefore, the final sizing was selected as a balanced solution that preserves gain, bandwidth, current consumption, input/output range, and correct operating

regions, while accepting the falling slew rate as the limiting large-signal performance metric.

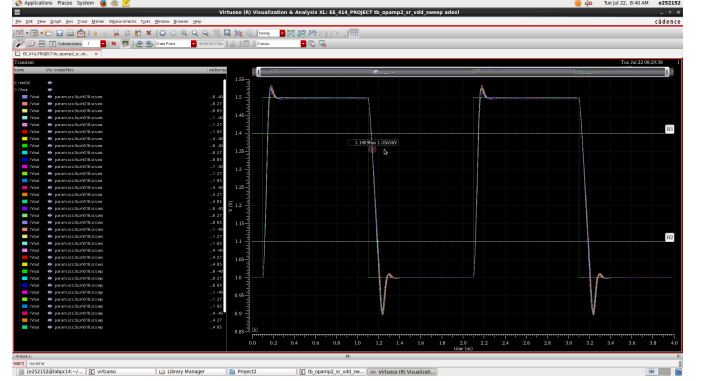


Fig. 4. Unity gain buffer slew rate simulation results of the operational amplifier.

E. Common-Mode Rejection Ratio

The common-mode rejection ratio (CMRR) was verified using an AC testbench. In this testbench, two identical op-amp instances were used to extract the differential-mode gain and the common-mode gain under the same simulation conditions. The CMRR was then calculated as the ratio between differential gain and common-mode gain:

$$CMRR = 20 \log_{10} \left(\frac{A_d}{A_{cm}} \right). \quad (2)$$

As shown in Fig. 5, the op-amp achieves around 100 dB CMRR in the nominal cases. Across the simulated PVT conditions, the worst-case CMRR is 93.97 dB, obtained at $V_{DD} = 3.3\text{ V}$, 85°C , SF corner. The best CMRR is obtained at $V_{DD} = 2.4\text{ V}$, -40°C , FF corner. Therefore, the op-amp satisfies the required 60 dB CMRR specification with a large margin in all tested cases.

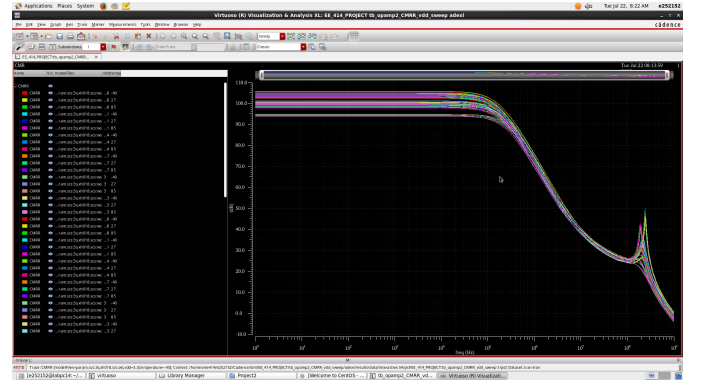


Fig. 5. CMRR simulation of the operational amplifier across PVT corners. The CMRR remains above the required 60 dB specification for all simulated cases.

TABLE III
OPERATIONAL AMPLIFIER CMRR VALUES

Condition	CMRR	Comment
Nominal cases	≈ 100 dB	Typical operation
Worst case	93.97 dB	$V_{DD} = 3.3$ V, 85°C , SF
Best case	105.86 dB	$V_{DD} = 2.4$ V, -40°C , FF

F. Input Common-Mode Range

The input common-mode range was verified using a unity-gain buffer testbench. In this configuration, the op-amp is expected to force the output to follow the input as long as the input pair and internal current-source devices remain in their proper operating regions. If the common-mode voltage exceeds the valid input range, the output starts to deviate from the input or the gain behavior becomes distorted.

For $V_{DD} = 1.8$ V, the required upper common-mode limit is $V_{DD} - 0.3$ V = 1.5 V. The simulation showed reasonable output tracking up to approximately 1.5 V. Above this level, around 1.6 V–1.7 V, the output behavior started to degrade due to limited voltage headroom. Since this region is outside the specified range for 1.8 V supply operation, the result is acceptable. The same test was repeated for higher supply voltages, including 3.3 V, where the output behavior was checked up to the corresponding upper common-mode limit of $V_{DD} - 0.3$ V. The unity-gain buffer simulations confirmed that the op-amp operates correctly across all simulated PVT cases within the specified input common-mode range.

G. Output Voltage Range

The output voltage range was verified using a level-shifted negative feedback testbench. The inverting input was connected to the midpoint of two equal $1\text{ M}\Omega$ resistors, one connected to V_{out} and the other to a control voltage V_{ctrl} . Therefore,

$$V_- = \frac{V_{out} + V_{ctrl}}{2}. \quad (3)$$

Since negative feedback forces $V_- \approx V_+$, the expected output voltage is

$$V_{out} = 2V_+ - V_{ctrl}. \quad (4)$$

A 10 pF load capacitor was connected at the output. By sweeping V_{ctrl} , different output voltage levels were forced and the simulated output was compared with the expected value. This test was repeated for all supply voltages.

TABLE IV
REPRESENTATIVE OUTPUT RANGE VERIFICATION

V_{DD}	V_+	V_{ctrl}	Expected V_{out}	Simulated V_{out}
1.8 V	1.0 V	1.0 V	1.0 V	≈ 1.0 V
1.8 V	1.0 V	1.4 V	0.6 V	≈ 0.6 V
3.3 V	1.5 V	0.5 V	2.5 V	≈ 2.5 V
3.3 V	1.5 V	2.4 V	0.6 V	≈ 0.6 V

The results show that the op-amp output follows the expected closed-loop relation within the required range of 0.6 V

to $V_{DD} - 0.8$ V. No significant degradation was observed within the specified output range for the simulated supply cases.

H. Design Tradeoffs

The main design tradeoff was between gain, bandwidth, slew rate, output swing, and current consumption. Larger transistor dimensions and longer channel lengths improve output resistance and DC gain, but they also increase parasitic capacitance and may reduce bandwidth. Higher bias current improves transconductance, unity-gain bandwidth, and slew rate, but increases total current consumption.

The selected topology and device sizing were therefore chosen to satisfy the required gain and bandwidth while keeping the supply current below the project limit. The final op-amp was verified through DC operating-point analysis, AC gain simulation, slew-rate simulation, CMRR simulation, input common-mode range simulation, output swing verification, and supply-current measurement.

IV. BANDGAP REFERENCE DESIGN

The bandgap reference circuit was based on the CMOS parasitic-PNP bandgap topology shown in Fig. 6. This topology is divided into three main blocks: a startup circuit, the bandgap core, and an internal amplifier. The core generates PTAT and CTAT components using two parasitic PNP devices with different emitter-area ratios, while the amplifier forces the proper equilibrium condition. The startup circuit is included to prevent the self-biased core from remaining in the zero-current state during power-up.

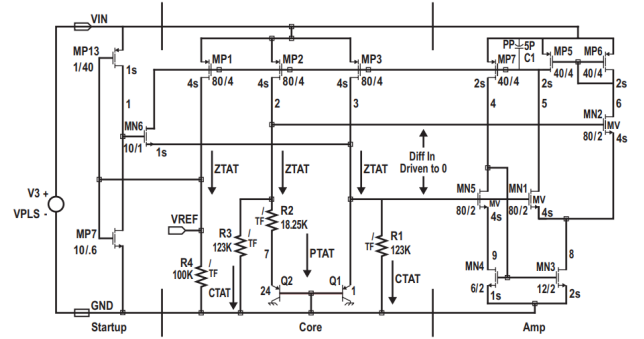


Fig. 6. Brokaw-based CMOS bandgap reference with startup, core, and amplifier block [3].

The main idea of the bandgap core is to combine a PTAT voltage with a CTAT voltage. The base-emitter voltage of a bipolar transistor decreases with temperature and therefore behaves as a CTAT quantity. On the other hand, the difference between two base-emitter voltages operated at different current densities is proportional to absolute temperature. This voltage is given by

$$\Delta V_{BE} = \frac{kT}{q} \ln(N), \quad (5)$$

where N is the emitter-area ratio. In this design, the parasitic PNP devices use an area ratio of approximately 24:1. Therefore, the generated PTAT voltage is proportional to $\ln(24)$.

The amplifier forces the two core branches to operate with nearly equal currents. Since the two PNP devices have different effective emitter areas, equal branch currents create different current densities and therefore generate ΔV_{BE} . This PTAT voltage appears across resistor R_2 . At the same time, resistors R_1 and R_3 create CTAT-related currents from the PNP emitter voltages. By properly selecting the resistor ratios, the PTAT and CTAT components are summed to obtain an approximately temperature-independent current.

A startup circuit was also included because the bandgap core is self-biased and can otherwise remain in the undesired zero-current state. During supply ramp-up, the startup circuit injects current into the core and helps the circuit move toward the correct operating point. After the reference voltage rises, the startup path turns off and no longer disturbs the normal bandgap operation. [3]

The DC operating point of the bandgap reference was also checked after the final sizing. As shown in Fig. 7, the circuit settles to a reference voltage close to 1 V while the internal bias nodes and branch currents remain properly established. This operating-point check confirms that the bandgap core, amplifier, and startup-related devices reach the intended bias condition and do not remain in the zero-current state.

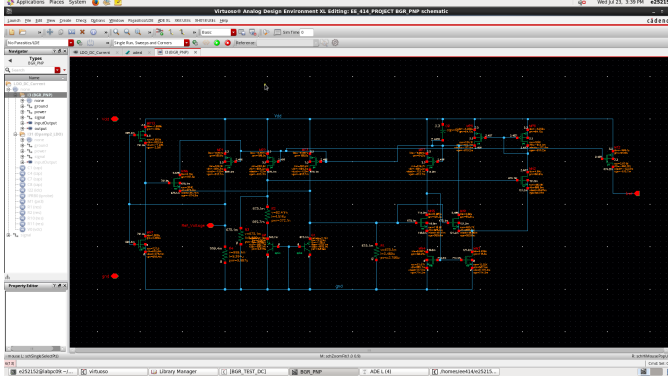


Fig. 7. DC operating point annotation of the bandgap reference.

A. Operating Temperature Range

The bandgap reference was simulated over the required temperature range from -40°C to 85°C . The simulation was repeated for different process corners and supply voltages to check whether the reference voltage remains stable under PVT variations. As shown in Fig. 8, the reference voltage stays close to 1 V over the full temperature range.

The minimum simulated reference voltage is approximately 0.993 V, while the maximum value is approximately 1.007 V. Therefore, the total variation is

$$\text{Variation} = \frac{1.007 - 0.993}{1.000} \times 100 \approx 1.4\%. \quad (6)$$

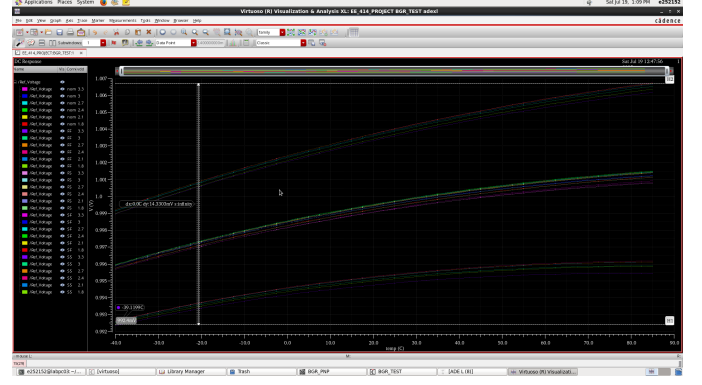


Fig. 8. Bandgap reference voltage over temperature and PVT corners.

TABLE V
BANDGAP TEMPERATURE RANGE RESULT

Parameter	Value
Temperature range	-40°C to 85°C
Minimum V_{REF}	$\approx 0.993\text{ V}$
Maximum V_{REF}	$\approx 1.007\text{ V}$
Total variation	$\approx 1.4\%$

Since the total variation is well below the required 5% limit, the bandgap reference satisfies the operating temperature range requirement.

B. Startup Operation

The startup operation of the bandgap reference was verified using a transient simulation with a ramped supply voltage. Since the bandgap core is a self-biased circuit, it can theoretically remain in the undesired zero-current state if no startup path is included. Therefore, the startup circuit is used to inject current into the core during power-up and force the circuit toward the correct operating point.

As shown in Fig. 9, the reference voltage initially remains close to zero while the supply voltage is still low. After the supply voltage becomes high enough, the startup circuit activates the core and the reference voltage rises toward its nominal value. Once the bandgap reaches the correct operating point, the startup path turns off and the reference settles close to 1 V. The simulation confirms that the circuit does not remain stuck at the zero-current state.

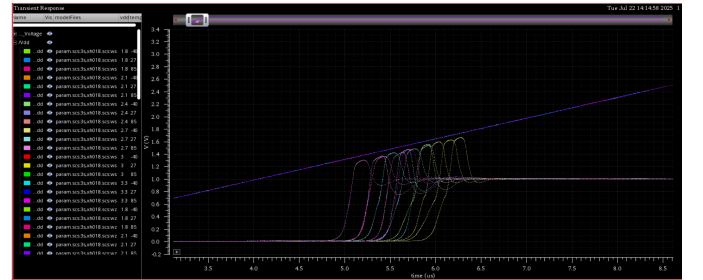


Fig. 9. Startup transient response of the bandgap reference.

V. LDO DESIGN

The low-dropout regulator (LDO) was designed using the previously developed operational amplifier and bandgap reference. The bandgap reference provides an approximately 1 V stable reference voltage, while the op-amp is used as the error amplifier of the regulator. The LDO output voltage is generated by comparing the bandgap reference voltage with a scaled version of the output voltage through a resistive feedback network.

A PMOS pass transistor is used as the main regulating device. Its source is connected to the input supply, its drain is connected to the regulated output, and its gate is driven by the op-amp output. When the output voltage decreases, the feedback voltage also decreases, and the op-amp adjusts the PMOS gate voltage to increase the pass current. Similarly, when the output voltage increases, the op-amp reduces the pass current. In this way, negative feedback keeps the output voltage close to the desired value. The implemented LDO topology is shown in Fig. 10.

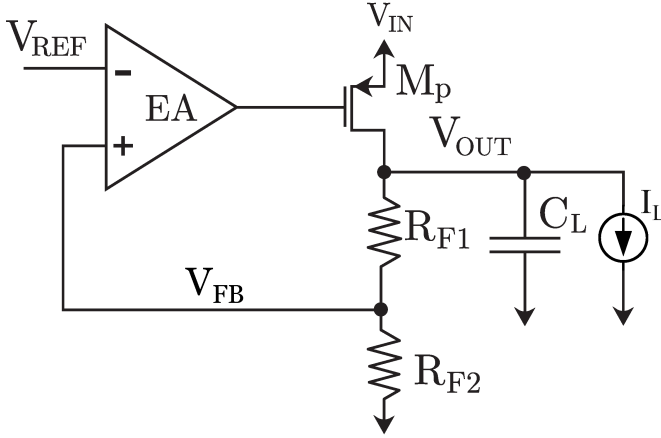


Fig. 10. Simplified LDO feedback topology.

The output voltage is set by the feedback resistor ratio as

$$V_{OUT} = V_{REF} \left(1 + \frac{R_1}{R_2} \right). \quad (7)$$

In the ideal case, $R_1 = 80 \text{ k}\Omega$ and $R_2 = 100 \text{ k}\Omega$ would generate exactly 1.8 V from a 1 V reference. However, in the final design, the feedback ratio was slightly adjusted to $R_1 = 79.5 \text{ k}\Omega$ and $R_2 = 100 \text{ k}\Omega$. This gives a nominal simulated output voltage of approximately 1.794 V. This small reduction was intentionally accepted to improve the regulator operating point and phase-margin behavior near the low supply condition. A further reduction of the output voltage could have improved the PSRR result, but it would move the regulated output farther away from the 1.8 V target. Therefore, 1.794 V was selected as a compromise between output accuracy, stability, and PSRR performance.

In addition to process, temperature, and supply-voltage variations, the LDO was verified for load capacitance values between 5 pF and 50 pF, and load-current values between 10 μA and 200 μA .

A. Desired Output Voltage

The LDO output voltage was verified under the simulated process, temperature, supply-voltage, load-current, and load-capacitance cases. The purpose of this test was to check whether the regulator output remains close to the desired 1.8 V level under DC operating conditions.

As shown in Fig. 11, the output voltage remains close to 1.8 V for the simulated cases. The nominal output voltage is approximately 1.794 V, which is slightly lower than the ideal target due to the final feedback resistor ratio selected as a compromise between output accuracy, stability, and PSRR performance. Across the tested cases, no DC regulation failure is observed.

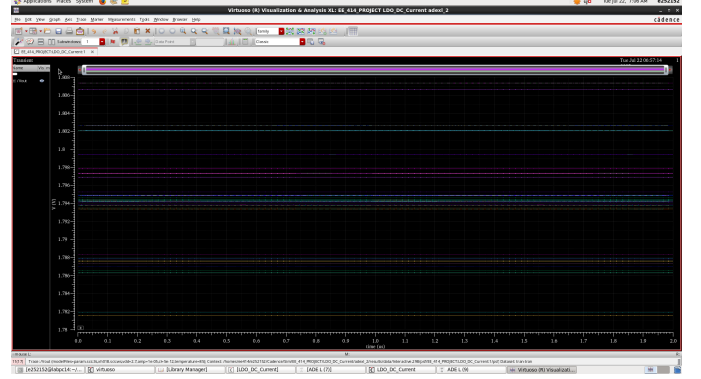


Fig. 11. LDO output voltage under DC operating conditions.

B. Total Supply Current

The total supply current of the LDO was simulated including the bandgap reference and the op-amp bias current. The test was repeated across the simulated process corners, temperature values, supply voltages, load capacitance values, and load-current cases.

As shown in Fig. 12, the total current remains below the 400 μA specification in every simulated case. Therefore, the LDO satisfies the total supply-current requirement with the bandgap reference included.

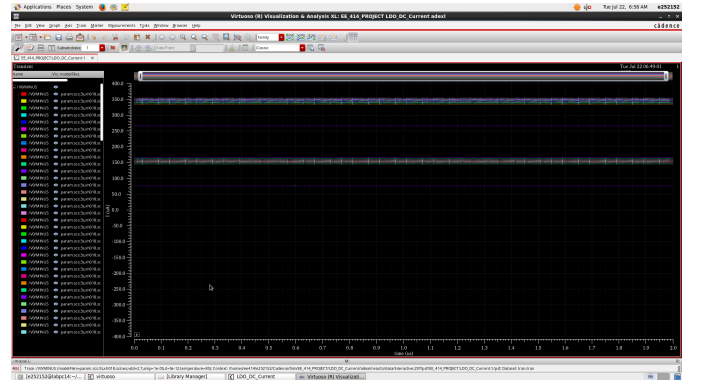


Fig. 12. Total LDO supply current including the bandgap reference.

The DC operating point of the complete LDO was also verified with the bandgap reference connected to the error

amplifier. As shown in Fig. 13, the bandgap output is close to 1 V, the feedback node tracks the reference input, and the regulated output settles around 1.794 V. This confirms that the op-amp, PMOS pass device, feedback divider, and load are correctly biased in the final closed-loop LDO configuration.

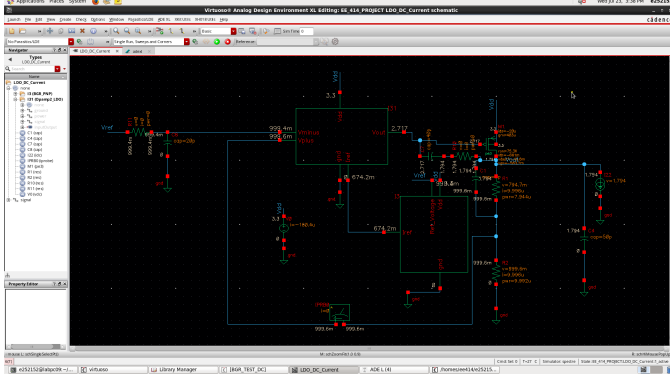


Fig. 13. DC operating point annotation of the complete LDO.

C. Instantaneous Load Current and Voltage Regulation

The instantaneous load current requirement was tested by applying a short triangular load-current pulse at the LDO output. The pulse reaches 50 mA with a total duration of 5 ns. Due to the very short pulse duration, an immediate voltage disturbance can occur at the output before the feedback loop fully reacts. Therefore, the important point in this test is whether the LDO output recovers and remains within the allowed regulation limit after the load transient.

Fig. 14 shows the recovered output voltage after the instantaneous load-current event for the simulated cases. The output settles close to the desired 1.8 V level, and the remaining deviation is lower than the allowed 150 mV transient regulation limit. Therefore, the LDO satisfies the instantaneous load-current requirement based on the settled output regulation after the transient event.

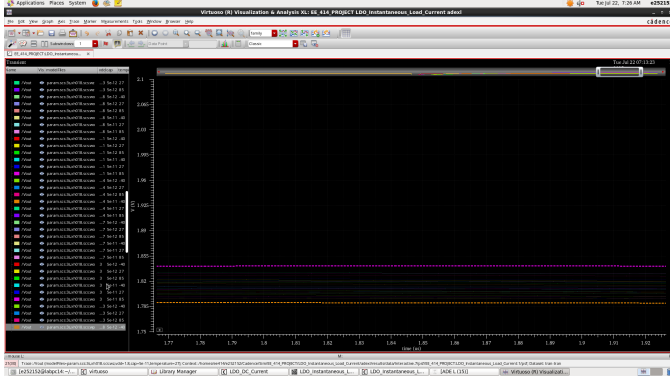


Fig. 14. Settled LDO output after the instantaneous load current test.

The minimum settled output occurs for $C_L = 5$ pF, -40°C , $V_{DD} = 1.8$ V, $I_L = 10$ μA , FF corner. The maximum settled output occurs for $C_L = 50$ pF, 85°C , $V_{DD} = 1.8$ V, $I_L = 200$ μA , SS corner. The worst recovered output deviation is

TABLE VI
SETTLED OUTPUT AFTER INSTANTANEOUS LOAD TEST

Case	Settled V_{OUT}
Minimum settled output	1.790 V
Maximum settled output	1.840 V
Worst deviation from 1.794 V	46 mV

therefore 46 mV, which is well below the 150 mV transient regulation limit.

D. PSRR

The PSRR of the LDO was tested by applying an AC signal to the supply voltage and observing the resulting AC component at the regulated output. In the simulation, the supply source was given an AC magnitude of 1 V, and the supply to output gain was evaluated at 10 kHz as

$$PSRR_{gain} = 20 \log_{10} \left(\frac{V_{out}}{V_{DD}} \right). \quad (8)$$

A lower supply to output gain means better supply rejection. Without any additional filtering between the bandgap reference and the error amplifier input, the LDO achieved approximately -47 dB from DC up to 10 kHz in most simulated cases. This value was close to the target but did not fully satisfy the -50 dB requirement over the specified frequency range.

The most problematic supply condition was $V_{DD} = 1.8$ V. Since the desired output voltage is also close to 1.8 V, the LDO operates near the dropout boundary at this supply voltage. In this condition, there is almost no voltage headroom across the PMOS pass transistor, and the error amplifier drives the pass device close to its limit. Therefore, the feedback loop cannot reject supply variations effectively, and PSRR degradation at 1.8 V is expected.

To improve the PSRR, a simple RC low-pass filter was added between the bandgap reference output and the op-amp reference input. The filter was implemented using a 7.5 M Ω resistor and a 20 pF capacitor. With this filter, the supply noise reaching the reference input was reduced, and the simulated PSRR improved to approximately -60 dB to -67 dB from DC up to 10 kHz across the valid regulated operating cases.

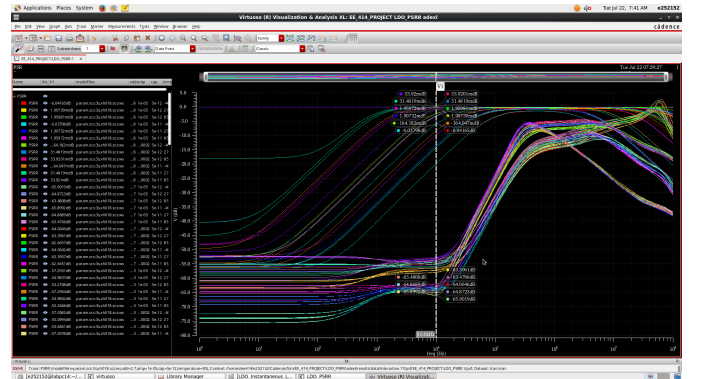


Fig. 15. LDO PSRR with the 7.5 M Ω –20 pF reference input filter.

E. Phase Margin

The phase margin of the LDO was one of the most challenging specifications in the project. The stability of the LDO depends not only on the op-amp itself, but also on the PMOS pass transistor, feedback network, load capacitance, and load-current condition. Therefore, satisfying all PVT, load-current, and load-capacitance cases at the same time was difficult.

The phase margin was measured using an STB loop-stability testbench in Cadence. An iprobe element was inserted in the feedback loop between the op-amp output and the PMOS pass transistor gate. Then, an STB analysis was performed to extract the loop gain, unity-loop-gain frequency, and phase margin without breaking the DC operating point of the regulator. This method allows the LDO loop stability to be evaluated while the circuit remains biased in its normal closed-loop operating condition.

During the design process, different compensation approaches were tested. First, a capacitor was placed in parallel with the feedback resistor to improve the loop response. Then, the internal Miller compensation of the op-amp was tuned to obtain a more stable LDO loop. The Miller compensation capacitor and its series nulling resistor are especially important because they control the dominant pole and the compensation zero of the op-amp. By increasing the Miller capacitor or by adjusting the nulling resistor zero, the op-amp phase margin can be increased further, which is a possible improvement for the most difficult LDO cases. The capacitance budget was also considered during this compensation tradeoff. Without further increasing the op-amp Miller compensation capacitor, the total capacitance used in the system is approximately 112.5 pF, including the load capacitances. If the Miller compensation capacitor is increased as an additional phase-margin improvement, about 36 pF more capacitance is added. In both cases, the total capacitance remains below the 200 pF project limit, but increasing the compensation capacitor reduces the remaining capacitance margin.

The phase-margin target was 60°. Most regulated operating cases satisfy this requirement. However, the 1.8 V supply cases are the most problematic, especially in the SS corner at 27 °C and 85 °C. This is mainly due to the limited voltage headroom: the desired output voltage is close to 1.8 V, so the LDO operates near the dropout boundary. In this condition, the PMOS pass device and the error amplifier have very limited room to regulate, and the loop-gain/phase-margin result becomes more sensitive and harder to satisfy.

The lowest passing phase margin was 61.35°, obtained at the FF corner, −40 °C, $V_{DD} = 2.7$ V, $I_L = 200$ μ A, and $C_L = 5$ pF. The best phase margin was 93.23°, obtained at the FF corner, 27 °C, $V_{DD} = 3.3$ V, $I_L = 10$ μ A, and $C_L = 5$ pF.

TABLE VII
LDO PHASE-MARGIN SUMMARY

Case	Phase Margin
Lowest passing case	61.35°
Best case	93.23°

VI. SIMULATION METHODOLOGY

All simulations were performed in Cadence Virtuoso using Spectre. Separate testbenches were used for the operational amplifier, bandgap reference, and LDO, and the final system was verified with the bandgap reference connected to the LDO.

The op-amp was characterized using DC operating-point, AC, transient, CMRR, input common-mode range, output range, and current-consumption simulations. The bandgap reference was verified using temperature, supply-voltage, process-corner, and startup simulations. The LDO was tested for desired output voltage, total supply current, instantaneous load-current response, PSRR, and phase margin.

The simulations covered TT, FF, SS, FS, and SF process corners, temperatures from −40 °C to 85 °C, and supply voltages from 1.8 V to 3.3 V. For the LDO, load capacitance values between 5 pF and 50 pF, and load-current values between 10 μ A and 200 μ A were also included.

VII. RESULTS AND DISCUSSION

The designed op-amp, bandgap reference, and LDO were verified separately and then evaluated together as a complete regulator system. Most of the required specifications were satisfied across the simulated process, voltage, temperature, load-current, and load-capacitance cases. The main design difficulty was the LDO stability and PSRR tradeoff near the low-supply condition.

A. Operational Amplifier Results

The operational amplifier satisfied the main low-power and small-signal requirements with a large margin. The total op-amp current remained below the 100 μ A limit in all simulated cases, with values approximately between 79 μ A and 94 μ A. The DC gain remained much higher than the required 40 dB specification, with a worst-case value of 97.34 dB. The unity-gain bandwidth also stayed above the 5 MHz target, with a minimum value of 5.787 MHz.

The CMRR requirement was also satisfied. The nominal CMRR was around 100 dB, and the worst-case value was 93.97 dB, which is well above the required 60 dB. The input common-mode range and output voltage range were verified across the required supply range, and no significant degradation was observed inside the specified operating ranges.

The main limitation of the op-amp was the falling slew rate. The rising slew rate was approximately 10.62 V μ s^{−1}, which satisfies the 8 V μ s^{−1} target. The falling slew rate was approximately 5.45 V μ s^{−1}, making it the limiting large-signal metric. This sizing was still kept because the op-amp satisfied the gain, bandwidth, current, CMRR, input/output range, and saturation-region requirements with good margin.

B. Bandgap Reference Results

The bandgap reference showed stable operation over the required temperature, supply, and process range. The simulated reference voltage stayed close to 1 V, with a minimum value

of approximately 0.993 V and a maximum value of approximately 1.007 V. This corresponds to a total variation of about 1.4 %, which is below the required 5 % limit.

The startup operation was also verified successfully. During supply ramp-up, the reference did not remain in the zero-current state and settled to its normal operating point close to 1 V. Therefore, the bandgap reference satisfies both the PVT variation and startup requirements.

C. LDO Results

The LDO generated an output voltage close to the desired 1.8 V target. The nominal output voltage was approximately 1.794 V. This value was selected as a compromise between output accuracy, stability, and PSRR performance. The output remained regulated across the tested process corners, temperatures, supply voltages, load capacitances, and load-current cases.

The total LDO supply current, including the bandgap reference and op-amp bias current, remained below the 400 μ A specification in all simulated cases. Therefore, the complete LDO system satisfies the total current requirement.

The instantaneous load-current test was evaluated from the recovered output voltage after the 50 mA, 5 ns load event. The settled output voltage stayed between approximately 1.790 V and 1.840 V. The worst recovered deviation from the 1.794 V target is therefore about 46 mV, which is well below the 150 mV transient regulation limit.

The phase-margin requirement was one of the most difficult parts of the design. After compensation tuning, the LDO achieved phase margin above 60° for the valid regulated operating cases. The lowest passing phase margin was 61.35°, while the best case was 93.23°. The 1.8 V supply condition remains the most sensitive case because the target output voltage is also close to 1.8 V, leaving very limited headroom across the PMOS pass device.

For PSRR, the unfiltered design achieved approximately -47 dB at 10 kHz, which was close to but not enough for the -50 dB target. After adding a 7.5 M Ω -20 pF RC filter between the bandgap reference and the op-amp reference input, the PSRR improved to approximately -60 dB to -67 dB at 10 kHz in the valid regulated cases. The 1.8 V supply case is still limited by dropout and headroom constraints, so it is treated separately from the normal regulated operating cases.

Overall, the complete design satisfies the main LDO requirements for output voltage, total current, recovered transient regulation, phase margin, and filtered PSRR in the regulated operating region. The most important remaining limitation is the reduced headroom at the lowest supply voltage, which affects loop stability and PSRR interpretation near dropout.

VIII. LAYOUT-AWARE DESIGN CONSIDERATIONS

Although layout was not required, the schematic design was considered with layout sensitivity in mind. The op-amp differential input pair should be placed symmetrically, preferably using common-centroid layout if possible. This reduces systematic mismatch and improves input offset and CMRR.

Current mirrors should use identical device dimensions, equal orientation, and symmetrical routing. Dummy devices should be added around matched transistor arrays to reduce edge effects.

For the bandgap reference, matching of resistor ratios is more important than absolute resistor values. The resistors that determine PTAT and CTAT weighting should be placed close together using common-centroid or interdigitated structures where possible. Thermal gradients should be minimized because they can introduce errors in the temperature-compensation behavior. The parasitic bipolar devices should also be placed symmetrically to improve matching.

For the LDO, the feedback node is sensitive because any noise or routing parasitic at this node directly affects output regulation. The feedback resistors should be placed close to the error amplifier input. The pass transistor should use wide metal routing for source and drain connections to reduce parasitic resistance. The gate routing of the pass device should be kept compact to limit parasitic capacitance that can degrade stability. Guard rings should be used around sensitive analog blocks to reduce substrate noise coupling. The total MIM capacitance must also remain below the project limit of 200 pF, so compensation capacitors and load-related capacitors must be selected carefully.

IX. CONCLUSION

In this project, an operational amplifier, a bandgap reference, and an LDO regulator were designed and simulated in the 180 nm CMOS process. The operational amplifier was first verified as a stand alone block and was then used as the error amplifier in the LDO system. The bandgap reference provided both the reference voltage and the bias current used in the final integrated design.

The operational amplifier satisfied the main small-signal requirements with a large margin. The supply current remained below the 100 μ A limit, the DC gain stayed above 97 dB, the unity-gain bandwidth remained above 5 MHz, and the CMRR was higher than 93 dB in all simulated cases. The input common-mode range and output voltage range were also verified successfully. The main limitation of the op-amp was the falling slew rate, which was lower than the target, while the rising slew rate satisfied the specification.

The bandgap reference generated an output close to 1 V over process, voltage, and temperature variations. The simulated reference voltage varied from approximately 0.993 V to 1.007 V, corresponding to about 1.4 % total variation. This is below the required 5 % limit. The startup simulation also confirmed that the circuit reaches the correct operating point after supply ramp-up.

The LDO generated a nominal output voltage of approximately 1.794 V. The total supply current, including the bandgap reference and op-amp bias current, remained below the 400 μ A specification. The instantaneous load-current test showed a recovered output deviation of about 40 mV, which is below the 150 mV transient regulation limit. The phase margin was also kept above 60° in the valid regulated operating cases.

With the added $7.5\text{ M}\Omega$ – 20 pF reference input filter, the PSRR improved to approximately -60 dB to -67 dB at 10 kHz .

Overall, the final design satisfies most of the project requirements. The most difficult part of the design was the LDO operation near the 1.8 V supply condition, where the desired output voltage is also close to 1.8 V . This creates limited headroom across the PMOS pass device and makes phase margin and PSRR more sensitive. Further improvement could be achieved by optimizing the compensation network, increasing the effective dropout headroom, or adding a more dedicated filtering and transient-assist structure.

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APPENDIX A CIRCUIT SCHEMATICS

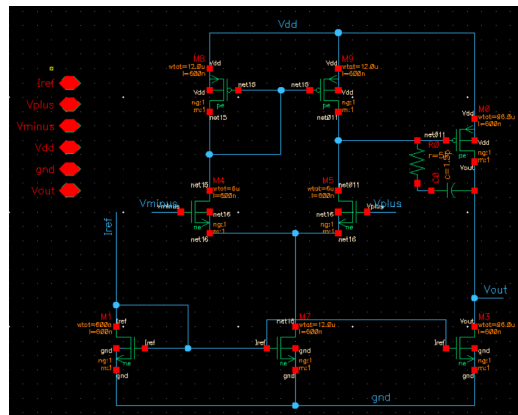


Fig. 16. Complete operational amplifier schematic.

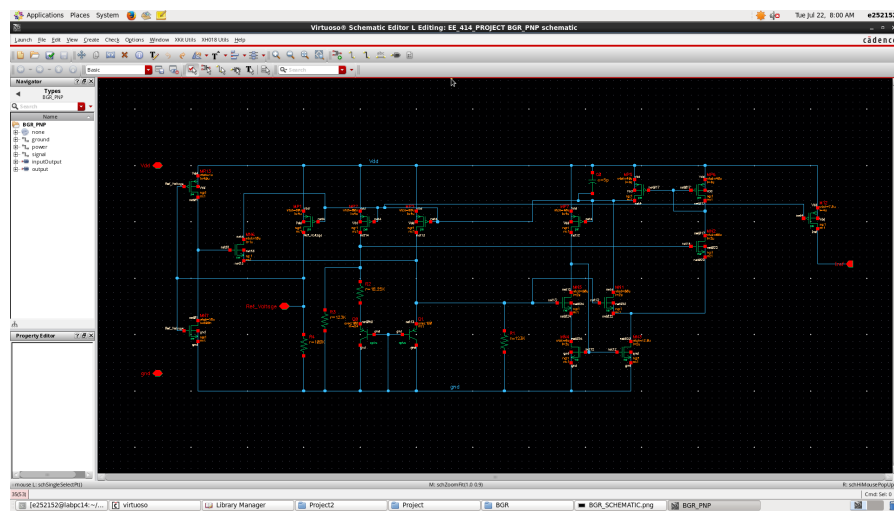


Fig. 17. Complete bandgap reference schematic.

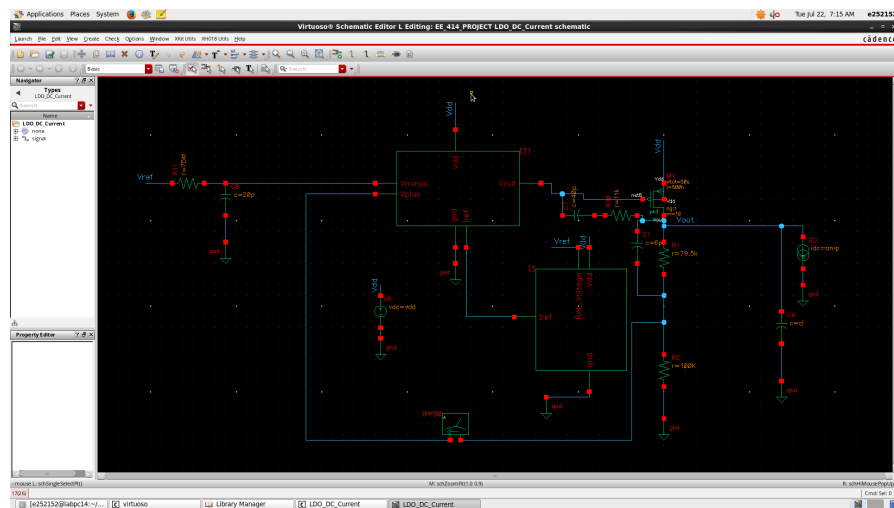
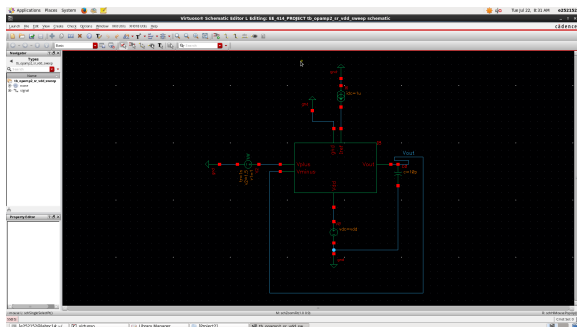
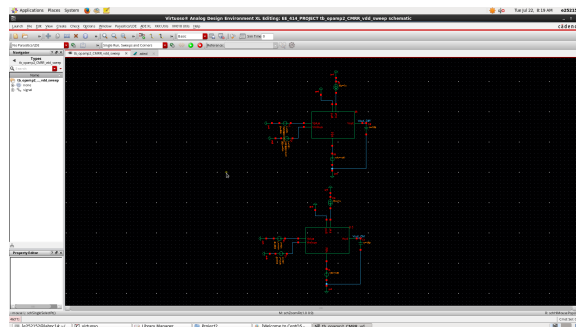


Fig. 18. Complete LDO schematic.

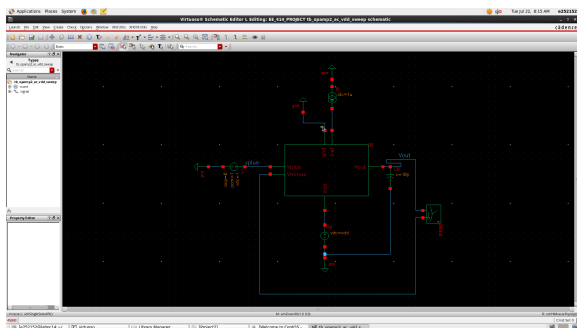
APPENDIX B SIMULATION TESTBENCHES



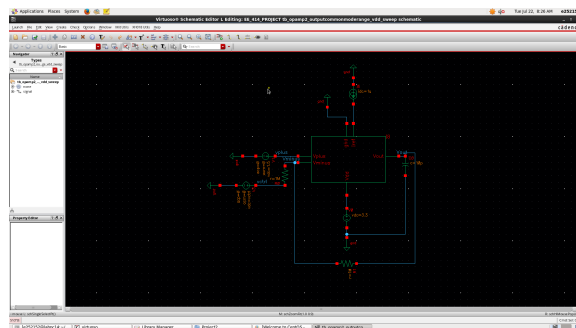
(a) Op-amp slew-rate testbench



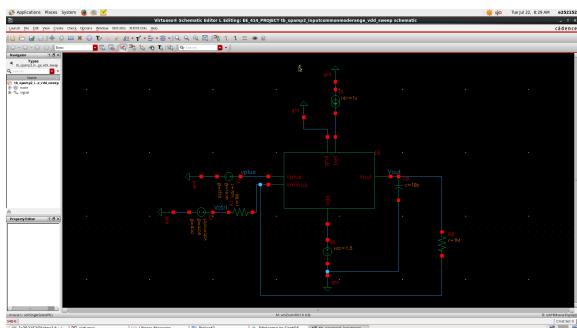
(b) Op-amp CMRR testbench



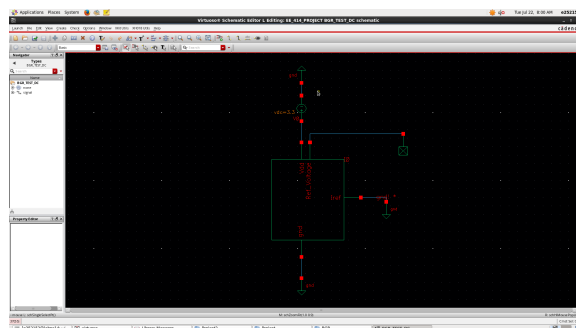
(c) Op-amp AC gain testbench



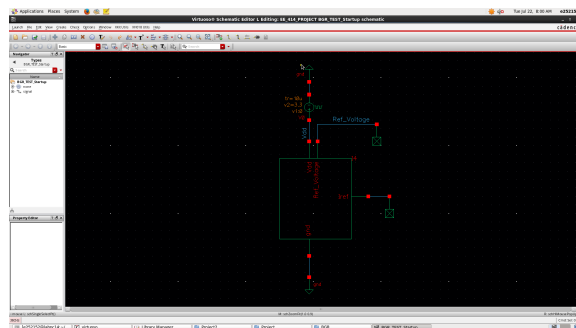
(d) Op-amp output range testbench



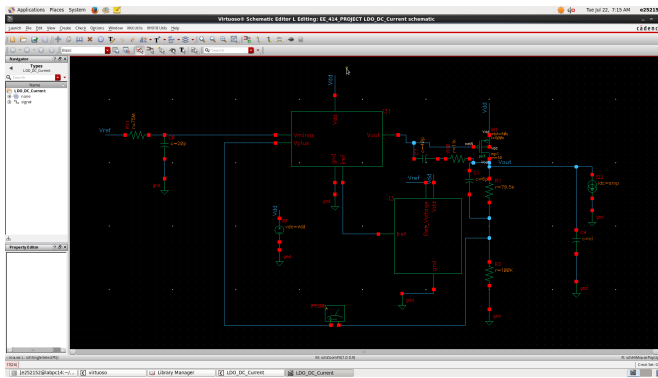
(e) Op-amp input common-mode range testbench



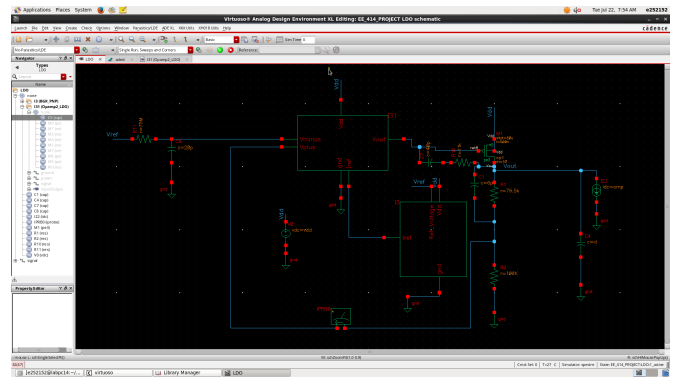
(f) Bandgap DC/PVT testbench



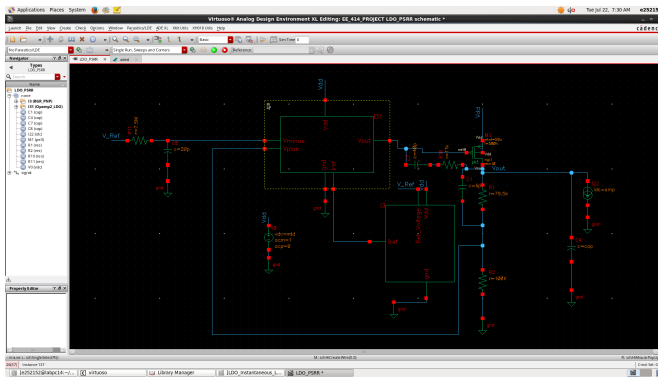
(g) Bandgap startup testbench



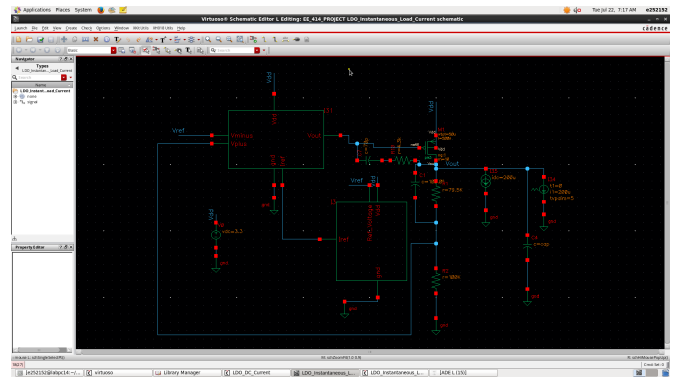
(a) LDO DC output and current testbench



(b) LDO phase-margin testbench



(c) LDO PSRR testbench



(d) LDO instantaneous load-current testbench